



Optical and radiative properties of Sm^{3+} ions activated alkali-bismuth-germanate glasses

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ABSTRACT

Trivalent samarium (Sm^{3+})-doped alkali-bismuth-germanate glasses of a composition $(40-x) \text{GeO}_2 + 20 \text{Bi}_2\text{O}_3 + 20 \text{Na}_2\text{O} + 10 \text{BaO} + 10 \text{Gd}_2\text{O}_3 + x \text{Sm}_2\text{O}_3$ ($x = 0.05, 0.1, 0.5, 1.0, 1.5$ and $2.5 \text{ mol}\%$) (GeBiNaBaGdSm) were made from a typical melt-quenching procedure. An optical absorption, photoluminescence excitation, emission and decay curves of GeBiNaBaGdSm glasses were studied. For the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition, an intense orange emission at 601 nm was observed when the Sm^{3+} ions pumped by 405 nm. A high value of $15.59 \times 10^{-22} \text{ cm}^2$ stimulated emission cross-section and $20.59 \times 10^{-28} \text{ cm}^3$ of optical gain bandwidth for the 1.5 mol% Sm_2O_3 -doped $\text{GeBiNaBaGdSm}15$ glass were obtained. The experimental lifetime (τ_{exp}) was increased up to $\text{GeBiNaBaGdSm}05$ glass, thereafter diminished with further increase of Sm^{3+} ion concentration. Decay profiles of Sm^{3+} ions were fitted using I-H model. Energy transfer rate (W_{ET}) for $\text{GeBiNaBaGdSm}10$ glass is evaluated to be 554 s^{-1} . The average red to orange (R/O) intensity ratio of these glasses is found to be 0.123. Furthermore, CIE chromaticity coordinates demonstrate that the emission was observed in the orange region and then shifted to red region with the increase of Sm^{3+} ion concentration. All the studies revealed that these glasses could be useful for tunable color display applications.

1. Introduction

Germanate glasses are of great attention for design and development of luminescent materials owing to their high refractive index, low phonon energies, and sensitivity to UV photon irradiation. Usually, germanate glasses have found applications in nonlinear optical devices, optical waveguides, Bragg gratings and optical fiber telecommunications [1–3]. With a significant addition of alkali metals ions such as Li^+ and Na^+ to these glasses which acts as mixed glass formers (MGF). The MGF contained glass types of electrolytes are exceptional candidates for next generation solid-state electrolytes. Besides, a better thermal stability and stronger mechanical strength of these glasses because of their sturdy inter-ionic force between Ge^{4+} and O^{2-} ions in comparison with

infrared (IR) transmitting glasses that include fluoride, tellurite and chalcogenide glasses [4].

Additionally, neutron scattering studies revealed that the $\text{GeO}_2 - \text{P}_2\text{O}_5$ and $\text{K}_2\text{O} - \text{GeO}_2 - \text{P}_2\text{O}_5$ glasses have been shown the existence of germanium atoms with a coordination number more than four. As a result, the structure of the GeO_2 has changed from tetrahedral to hexagonal. This higher coordinated germanium would cause a significant modifications in the structure of the binary alkali glasses, especially the alkali concentration less than 20 mol%, where nonlinear trends were noticed in the macroscopic properties that is said to be germinate anomaly [5–7]. The germanate glasses are superior in the optical fiber technology compared to silicate glasses due to their transparency in the mid infrared (mid-IR) region. But the development

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of these glasses was hindered by Rayleigh scattering losses [8]. These losses can be minimized with the addition of alkali metal ion modifiers in the germanate glass.

Among lanthanide (Ln^{3+}) ions, Gd_2O_3 is a widespread material because of proficient energy transfer occurs to the Ln^{3+} ions from Gd^{3+} ions consequently high thermal neutron capture cross-section that increases the light yield of emission. Addition of alkali metal ions (Li^+ , Na^+ , K^+) modifies, the Ge network leads to the formation of non-bridging (NBO) oxygens. The high Na^+ ion concentration may decrease ion pair function due to the result of bridging oxygen (BO) to non-bridging where Na–NBO distances are smaller in Na–O–Na bonds compared to Na–O–Bi/Ge that attributed to strong coulomb attractive Ge–O interaction. The addition of Na^+ also decreases defects in solids due to its smaller cation size [9–11].

Samarium (Sm^{3+}) ion is a good choice among the Ln^{3+} ions as its lowest emitting energy state $^4\text{G}_{5/2}$ possesses higher quantum efficiency with different quenching channels and exhibits a very small amount of probability for non-radiative decay, which are worthy characteristics of laser applications. Sm^{3+} ions activated glasses have realized in the promising applications as high density optical storage, color displays, solid state lasers (in visible region), photodynamic therapy (PDT) light sources and telecommunication (undersea) [12]. Sm^{3+} -doped glasses show a strong orange-red emission in the visible region. Sm^{3+} ions exhibit broad emission intensity bands in the NIR region due to $^4\text{G}_{5/2} \rightarrow ^6\text{H}_J$ transitions (where $J = 5/2, 7/2, 9/2$ and $11/2$). A large energy gap ($\sim 7250 \text{ cm}^{-1}$) between meta stable state $^4\text{G}_{5/2}$ and lower $^6\text{F}_{11/2}$ state in Sm^{3+} ions would cause a reddish orange emission around 600 nm which is not stimulated dreadfully by the phonon energy of the glass matrix [13–15]. For the purpose of energy transfer studies, Sm^{3+} ion can be used as co-doped ion with other Ln^{3+} ions [16–22].

This study is aimed to synthesize the Sm^{3+} -doped alkali bismuth-germanate ($\text{GeO}_2 + \text{Bi}_2\text{O}_3 + \text{Na}_2\text{O} + \text{BaO} + \text{Gd}_2\text{O}_3$, GeBiNaBaGd) glasses and to investigate their optical, photoluminescence and radiative properties with the variation of Sm^{3+} ion concentration. The radiative transition probabilities (A), branching ratio (β) and stimulated emission cross-sections (σ_{SE}) of GeBiNaBaGdSm15 glass were evaluated with the help of Judd-Ofelt (JO) theory. Inokuti and Hirayama (IH) model is employed to fit the non-exponential decay profiles of Sm^{3+} ions in these glasses, with an aim to know the multipolar interactions among Sm^{3+} ions and explore the energy transfer processes.

2. Glass preparation and characterization techniques

2.1. Glass fabrication procedure

A batch of 10 gm from high purity chemical composition of $(40-x)\text{GeO}_2 + 20\text{Bi}_2\text{O}_3 + 20\text{Na}_2\text{O} + 10\text{BaO} + 10\text{Gd}_2\text{O}_3 + x\text{Sm}_2\text{O}_3$ ($x = 0.05, 0.1, 0.5, 1.0, 1.5$ and 2.5) (GeBiNaBaGdSm) glasses were prepared using conventional melt-quenching technique. These glasses were coded as GeBiNaBaGdSm005 , GeBiNaBaGdSm01 , GeBiNaBaGdSm05 , GeBiNaBaGdSm10 , GeBiNaBaGdSm15 and GeBiNaBaGdSm25 . The stoichiometrical amounts of chemicals were mixed homogeneously by grinding all the high purity chemicals using agate mortar and pestle. The powder was melted in the alumina crucible at 1260°C for 30 min. Then the melt was decanted on a pre-heated 5 mm thick brass plate at 400°C . The obtained glasses were annealed at 400°C for 90 min to release the internal stresses in the glasses. Then the glasses were polished for the optical characterization. All composition reported are expected composition from the relative weights of the compounds used in the preparation of the glasses.

Optically polished GeBiNaBaGdSm glasses were used to derive their photoluminescence properties. Thickness of the glasses is measured using digital screw gauge and density of the glass is measured by Archimedes principle using Sartorius digital analytical balance (Germany). The physical properties (thickness, density and concentration) of the studied glasses are shown in Table 1 along with their

chemical composition. Refractive index (n) of a very thin and fine polished Sm^{3+} -doped GeBiNaBaGdSm15 glass was measured using J.A. Woollam Spectroscopic Ellipsometer (Model: M-2000VI) in the spectral range of 370–1690 nm with an error of ± 0.0023 . The refractive index of GeBiNaBaGdSm15 glass is found to be 1.5859 at a Sodium wavelength of 589 nm.

2.2. Spectroscopic characterization methods

Absorption spectrum of GeBiNaBaGdSm15 glass powder was recorded by using double beam UV-visible-NIR VARIAN Cary-5000 (175–3300 nm range) spectrophotometer in diffuse reflectance (DRS) mode. Photoluminescence (PL) excitation, emission and decay curves of the Sm^{3+} ions were measured by Edinburgh Spectrofluorometer with a spectral resolution of 0.01 nm (model: FLS – 980).

3. Results and discussion

3.1. Optical (UV- visible- NIR) absorption spectrum

The ultraviolet (UV) – visible – near infrared (NIR) optical absorption spectrum of GeBiNaBaGdSm15 glass was recorded in the 350–2500 nm spectral region at room temperature and is shown in Fig. 1. Twelve optical absorption bands from visible and NIR were observed in the absorption spectrum owing to $4f^5-4f^6$ transitions. Six absorption bands observed in the range of 400–500 nm are shown in Fig. 1(a), and the rest of the bands noticed in the NIR region in the range of 900–2200 nm for GeBiNaBaGdSm15 glass are shown in Fig. 1 (b). On the other hand, a transmission spectrum of the same glass is also presented in the inset of Fig. 1 (a). This reveals that the glass exhibits the transmission as high as about 90% in the visible and NIR regions. The quality of physical transparency of the GeBiNaBaGdSm glass is presented in the inset of Fig. 1(b). The transitions take place from the ground state $^6\text{H}_{5/2}$ to the higher energy levels assigned as $^4\text{F}_{5/2}$, $^4\text{I}_{13/2}$, $^4\text{I}_{11/2}$, $^4\text{I}_{9/2}$, $^6\text{F}_{11/2}$, $^6\text{F}_{9/2}$, $^6\text{F}_{7/2}$, $^6\text{F}_{5/2}$, $^6\text{F}_{3/2}$, $^6\text{H}_{15/2}$, $^6\text{F}_{1/2}$, $^6\text{H}_{13/2}$ levels. The corresponding wavelengths of the bands of Sm^{3+} ions with their energies were presented in Table 2. The absorption spectrum revealed two high intensity transitions assigned as $^6\text{H}_{5/2} \rightarrow ^6\text{F}_{7/2}$ and $^6\text{H}_{5/2} \rightarrow ^6\text{F}_{9/2}$ in the NIR region than those of the other transitions.

These transitions of optical absorption were initiated because of the electric-dipole influence which follows the selection rule, $\Delta J \leq 6$. The $^6\text{H}_{5/2} \rightarrow ^6\text{F}_{7/2}$ transition is due to the magnetic-dipole that follow the selection rule $\Delta J = 0, \pm 1$. The $^6\text{H}_{5/2} \rightarrow ^6\text{F}_{7/2}$ transition is found to be a hypersensitive transition among the observed absorption bands. The hypersensitive absorption band is because of the inhomogeneous polarizability of the glass which was a result of variation of the electric vector of the electromagnetic radiation around Sm^{3+} ion, thus significantly increases the intensities of quadrupole transitions.

3.2. Oscillator strengths

According to Judd – Ofelt (JO) theory [23,24], the local modifications in the glass structure around the Ln^{3+} ions can result in the variation of their optical and luminescence properties. With the use of integrated area of the absorption spectral bands and index of refraction (n), JO intensity parameters of the Sm^{3+} ions in GeBiNaBaGdSm15 glass were estimated. Table 2 presents the estimated values of oscillator strengths such as experimental (f_{exp}) and calculated (f_{cal}) strengths of the observed transitions. The value of f_{exp} of a specific spectral transition can be obtained from the integrating area of the optical absorption bands estimated from equation (1):

$$f_{\text{exp}} = 4.319 \times 10^{-9} \left(\frac{\text{mol} - \text{cm}^2}{l} \right) \int \epsilon(\nu) d\nu. \quad (1)$$

where ϵ is the molar absorptivity of the wavenumber ν (cm^{-1}).

Table 1

Comparison of glass composition, thickness and density of GeBiNaBaGdSm glasses.

S.No	Glass composition	Glass code	Thickness (mm)	Density (gm/cm ³)
1	39.95GeO ₂ + 20Bi ₂ O ₃ + 20Na ₂ O + 10BaO + 10Gd ₂ O ₃ + 0.05Sm ₂ O ₃	GeBiNaBaGdSm005	2.20	3.60
2	39.90GeO ₂ + 20Bi ₂ O ₃ + 20Na ₂ O + 10BaO + 10Gd ₂ O ₃ + 0.1Sm ₂ O ₃	GeBiNaBaGdSm01	1.49	3.41
3	39.50GeO ₂ + 20Bi ₂ O ₃ + 20Na ₂ O + 10BaO + 10Gd ₂ O ₃ + 0.5Sm ₂ O ₃	GeBiNaBaGdSm05	2.32	3.66
4	39.0GeO ₂ + 20Bi ₂ O ₃ + 20Na ₂ O + 10BaO + 10Gd ₂ O ₃ + 1.0Sm ₂ O ₃	GeBiNaBaGdSm10	2.38	3.67
5	38.5GeO ₂ + 20Bi ₂ O ₃ + 20Na ₂ O + 10BaO + 10Gd ₂ O ₃ + 1.5Sm ₂ O ₃	GeBiNaBaGdSm15	2.10	3.69
6	37.5GeO ₂ + 20Bi ₂ O ₃ + 20Na ₂ O + 10BaO + 10Gd ₂ O ₃ + 2.5Sm ₂ O ₃	GeBiNaBaGdSm25	3.02	3.72

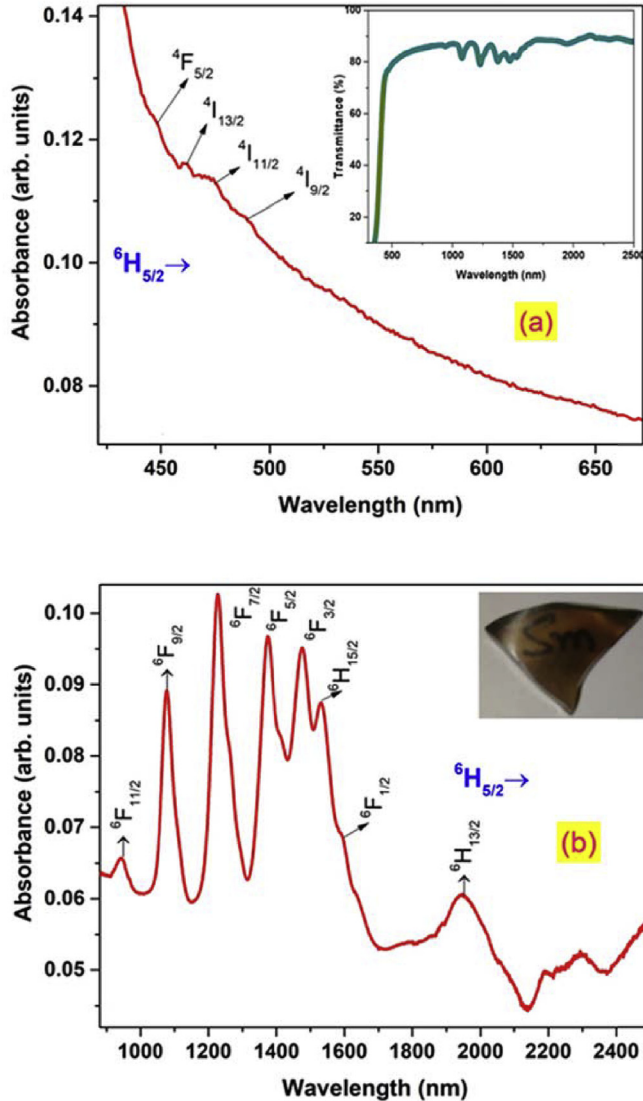


Fig. 1. Optical absorption spectrum of GeBiNaBaGdSm15 glass in (a) UV-visible (b) NIR regions. A transmission spectrum of the glass sample is presented in the inset of Fig. 1(a). Image and its optical quality presented in the inset of Fig. 1(b).

$\int \epsilon(\nu) d\nu$ represents the integrated area of the peak. In addition, f_{cal} is the induced electric dipole (ED) transition from ground level (ψJ) to the excited $4f$ level ($\psi' J'$). This f_{cal} was evaluated by the least square fitting process from Eq. (2). Execution of the JO theory, the intensity of three phenomenological parameters Ω_2 , Ω_4 and Ω_6 can be evaluated by using Eq. (2).

$$f_{\text{cal}} = \frac{8\pi^2 m c \nu}{3h(2J+1)} \frac{(n^2+2)^2}{9n} \chi_{\text{ed}} \sum_{\lambda=2,4,6} \Omega_{\lambda} (\psi J || U^{\lambda} || \psi' J')^2 \quad (2)$$

where m is the mass of the electron, c is the velocity of light, ν is the wavenumber of a particular transition, n is the refractive index of the glass, h is the Planck's constant, J is the total angular momentum of the ground state; Ω_2 , Ω_4 & Ω_6 are the JO intensity parameters and $||U^{\lambda}||$ are matrix elements of a unit tensor operator. In addition, the transition $^4G_{5/2} \rightarrow ^6H_{7/2}$ generally represents hypersensitivity nature because of the high field strength. The small value of δ_{RMS} (0.075) indicates a good correlation between f_{exp} and f_{cal} .

The nephelauxetic ratio (β) and bonding parameters (δ) are used to know about the nature of the Sm^{3+} and ligand bond in the glass environment. The nephelauxetic ratio is defined as the ratio of ν_c to ν_a which are the wavenumbers of the Sm^{3+} transitions of GeBiNaBaGdSm15 glass and for the aqua-ion [25], respectively. The bonding parameter has been obtained by following procedure described elsewhere [26]. Based on sign of value of the δ , ionic or covalent bond of the Ln^{3+} ion – ligand can be predicted. The negative sign of δ (-0.021) for GeBiNaBaGdSm15 glass indicates ionic bond character of Sm^{3+} ions ligand.

Spectroscopic quality factor (χ) ($= \Omega_4/\Omega_6$) is an essential factor to anticipate the behaviour of several lasing transitions of Sm^{3+} ions in the glass matrix. χ value for GeBiNaBaGdSm15 glass is found to be 0.67 which is greater than 0.52 for (60-x) TeO_2 -20 WO_3 -15 PbO -5 ZnO -x Sm_2O_3 (ZnPBWTe10) tellurite glass [34] but less compared to 0.74 for (80-x) NaPO_3 – 20 NaF – x Sm_2O_3 (NNS05) phosphate – glass [33], 1.48 for (70-x) B_2O_3 -15 Na_2O -15 K_2O -x Sm_2O_3 (NKBSm10) borate glass [29], 1.61 for (60-x) P_2O_5 -20 MgO -20 ZnSO_4 -x Sm_2O_3 (PZSMSm15) phosphate glass [28], 1.75 for (56-x/2) P_2O_5 -17 K_2O - (12-x/2) BaO - 6 BaF_2 -9 Al_2O_3 - x Sm_2O_3 (PKBFAS10) phosphate glass [26], 1.77 for 10- ZnO :10 Al_2O_3 :20 BaO : (60-x) B_2O_3 : x Sm_2O_3 (ZABBSm03) borate glass [35], 1.79 for 3 CdO - Al_2O_3 -3 SiO_2 (CASSm05) silicate glass [36], 1.93 for 20 PbO - 10 Al_2O_3 - 10 ZnO -59 B_2O_3 - 1 Sm_2O_3 (ZPABSm10) borate glass [30], 2.04 for 30 B_2O_3 -40 GeO_2 -(30-x) Gd_2O_3 -x Sm_2O_3 (BGGSm10) germanate glass [27], 2.22 for 49 PbO -30 GeO_2 -20 TeO_2 (LGTSm10) germanate glass [32] and 2.71 for (40-x) SiO_2 -20 Nb_2O_5 -20 K_2O -20 ZnF_2 -x Sm_2O_3 (SNbKZFSm10)silicate glass [31]. JO intensity parameters, quality factor and its trend of GeBiNaBaGdSm15 glass with various glasses are presented in Table 3.

4. Photoluminescence (PL) excitation spectra

The PL excitation spectra of the Sm^{3+} -doped GeBiNaBaGdSm glasses were recorded by monitoring the emission at 601 nm in the range of 350–540 nm at room temperature and are shown in Fig. 2. The spectra consists of twelve excitation bands positioned at 361 nm, 374 nm, 389 nm, 401 nm, 414 nm, 420 nm, 437 nm, 461 nm, 471 nm, 489 nm, 500 nm and 525 nm are ascribed to the transitions from the ground state, $^6H_{5/2}$ to $^4D_{3/2}$, $^6P_{7/2}$, $^4L_{5/2}$, $^6P_{3/2}$, $^6P_{5/2}$, $^4M_{19/2}$, $^4G_{9/2}$, $^4I_{13/2}$, $^4I_{11/2}$ + $^4I_{9/2}$, $^4M_{15/2}$, $^4G_{7/2}$ and $^4F_{3/2}$ states, respectively. Inner $4f^5$ - $4f^6$ electronic transitions are responsible for these excitation bands of Sm^{3+} ions. Among all the bands, highest excitation intensity was perceived for the transition $^6H_{5/2} \rightarrow ^6P_{3/2}$ centred at 401 nm. The variation of excitation intensity as a function of Sm^{3+} ions concentration is shown in the inset of Fig. 2. The GeBiNaBaGdSm01 glass unveils intense absorption compared to other Sm^{3+} ions doped GeBiNaBaGdSm glasses. There is a decrease in absorption of the glasses at high Sm^{3+}

Table 2

Observed optical absorption band positions (nm) and their energies (cm^{-1}), oscillator strengths (f_{exp} & $f_{\text{cal}} \times 10^{-6}$) of Sm^{3+} ions in GeBiNaBaGdS15 glass for different transitions.

Wavelength (λ) nm	Energy (ν) cm^{-1}	Transitions from $^6\text{H}_{5/2}$	f_{exp}	f_{cal}	$ \Delta f = f_{\text{exp}} - f_{\text{cal}} $
449	22272	$^4\text{F}_{5/2}$	0.0701	0.0035	0.0666
460	21739	$^4\text{I}_{13/2}$	0.0501	0.0032	0.0469
474	21097	$^4\text{I}_{11/2}$	0.0765	0.0114	0.0651
489	20449	$^4\text{I}_{9/2}$	0.1997	0.0081	0.1916
942	10616	$^6\text{F}_{11/2}$	0.1296	0.0238	0.1058
1073	9319	$^6\text{F}_{9/2}$	0.1496	0.2131	0.0635
1227	8149	$^6\text{F}_{7/2}$	0.2215	0.1885	0.0330
1372	7288	$^6\text{F}_{5/2}$	0.0837	0.0745	0.0092
1476	6775	$^6\text{F}_{3/2}$	0.0458	0.0559	0.0101
1529	6540	$^6\text{H}_{15/2}$	0.0338	0.0013	0.0325
1559	6414	$^6\text{F}_{1/2}$	0.0367	0.0340	0.0027
1947	5136	$^6\text{H}_{13/2}$	0.0664	0.0154	0.0510

$$\beta = 1.022,$$

$$\delta = -0.021 \pm 0.001,$$

$$\delta_{\text{RMS}} = \pm 0.01 (12)$$

Table 3

Comparison of JO intensity Ω_{λ} ($\times 10^{-20} \text{cm}^2$) parameters, their trend and quality factor (χ) of Sm^{3+} glasses.

Glass matrix	Ref.	Ω_2	Ω_4	Ω_6	χ	Trend
GeBiNaBaGdSm15	[PW]	0.28	0.32	0.48	0.67 ± 0.03	$\Omega_6 > \Omega_4 > \Omega_2$
PKBFAS10	[26]	3.46	5.75	3.28	1.75	$\Omega_4 > \Omega_2 > \Omega_6$
BGGSm10	[27]	1.66	2.89	1.42	2.04	$\Omega_4 > \Omega_2 > \Omega_6$
PZSmsm15	[28]	5.32	4.70	2.91	1.61	$\Omega_2 > \Omega_4 > \Omega_6$
NKBsm10	[29]	6.46	2.89	2.19	1.48	$\Omega_2 > \Omega_4 > \Omega_6$
ZPABsm10	[30]	2.36	4.84	2.51	1.93	$\Omega_4 > \Omega_6 > \Omega_2$
SNbKZFSm10	[31]	1.19	6.71	2.48	2.71	$\Omega_4 > \Omega_6 > \Omega_2$
LGTsm10	[32]	0.73	4.56	2.05	2.22	$\Omega_4 > \Omega_6 > \Omega_2$
NNS05	[33]	0.49	6.58	8.89	0.74	$\Omega_6 > \Omega_4 > \Omega_2$
ZnPbWTe10	[34]	0.40	3.08	5.94	0.52	$\Omega_6 > \Omega_4 > \Omega_2$
ZABBSm03	[35]	4.58	5.62	3.17	1.77	$\Omega_4 > \Omega_2 > \Omega_6$
CASSm05	[36]	2.87	3.34	1.86	1.79	$\Omega_4 > \Omega_2 > \Omega_6$

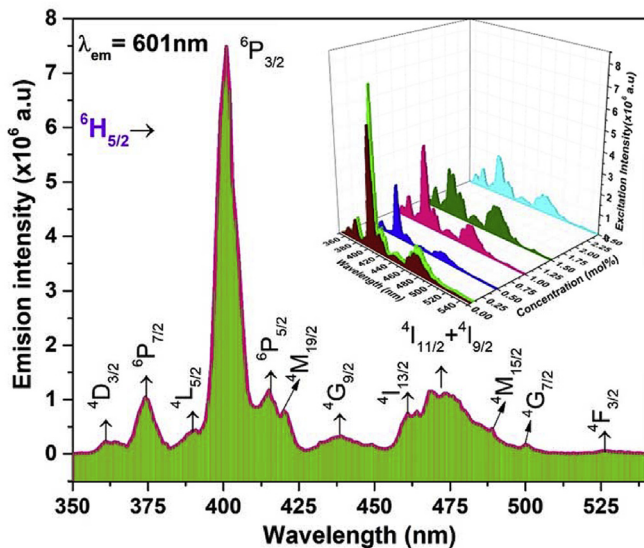


Fig. 2. Excitation spectrum of GeBiNaBaGdSm01 glass attained by monitoring the emission wavelength at 601 nm. Inset shows the excitation spectra of different GeBiNaBaGdSm glasses.

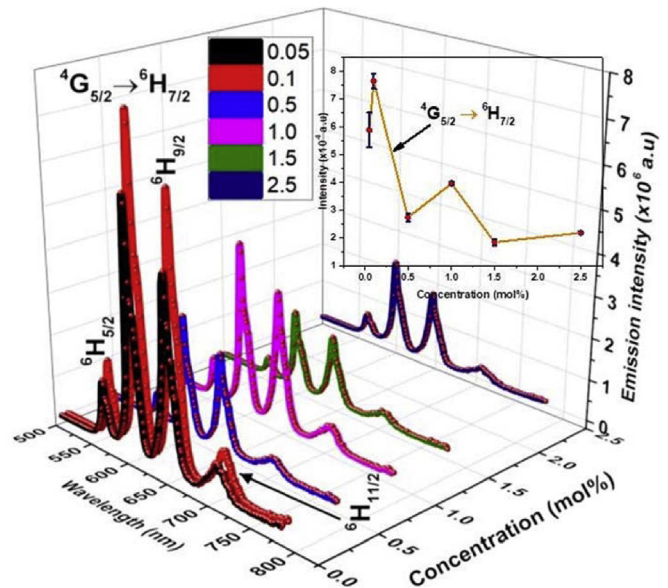


Fig. 3. Emission spectra of GeBiNaBaGdSm glasses upon 405 nm excitation. Inset shows the variation of orange emission intensity as a function of Sm^{3+} ion concentration.

ion concentrations. The similar trend was attained in the emission spectra of these glasses.

5. PL spectra

PL spectra of GeBiNaBaGdSm glasses with variation of Sm^{3+} ion concentration were recorded upon excitation of 405 nm in the spectral range of 530–740 nm, as shown in Fig. 3. The spectra revealed four emission bands of yellow, orange, orange-red and red were positioned at 561 nm (17825 cm^{-1}), 597 nm (16750 cm^{-1}), 644 nm (15527 cm^{-1}) and 707 nm (14144 cm^{-1}), which are ascribed to $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{5/2}$, $^6\text{H}_{7/2}$, $^6\text{H}_{9/2}$ and $^6\text{H}_{11/2}$ transitions, respectively. Similar emissions were noticed in the reported germanate glasses [37,38]. Among these, the intense emission band at 597 nm was observed with a FWHM of 13.21 nm. This transition is due to a partially magnetic dipole (MD) and electric dipole (ED) obeying the selection rule of $\Delta J = \pm 1$. As the Sm^{3+} ion concentration increases, the emission intensity was also increased up to 0.1 mol% of Sm^{3+} ions concentration and then the intensity decreases as the concentration increases with small fluctuations. Error bars were also included for the orange emission intensity as a

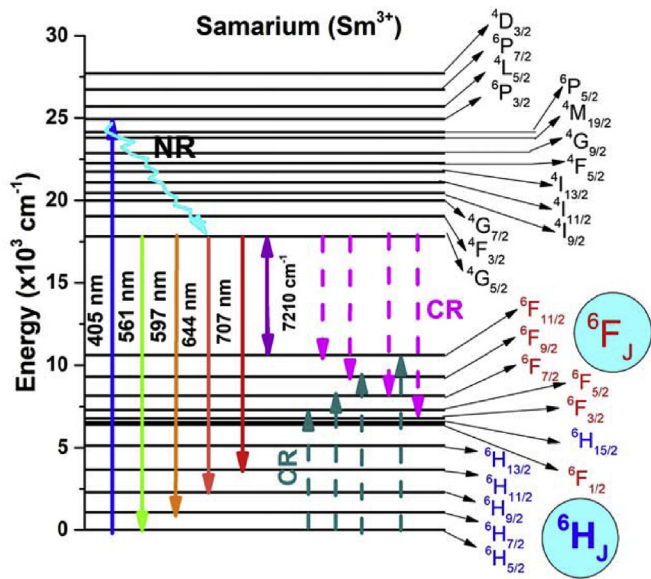


Fig. 4. Energy level diagram of Sm^{3+} ions in the GeBiNaBaGdSm glasses, with various transition levels. Excitation, radiative and non-radiative transitions also shown along with cross-relaxation (CR) channels.

function of Sm^{3+} ion concentration to know the maximum and minimum values, as shown in the inset of Fig. 3. The quenching of luminescence is due to the clustering and also energy transfer of Sm^{3+} ions at higher concentration. The excitation spectra revealed a similar behaviour with increase of Sm^{3+} ions concentration that was explained in the previous section. Moreover, photoluminescence properties depend on the geometry and size of the materials doped with RE^{3+} ions where strong overlap of emission and absorption bands of RE^{3+} ions existed, leads to change in shape of the emission spectra. It is clearly observed from excitation (Fig. 2) and emission (Fig. 3) spectra of the Sm^{3+} -doped glasses that there is no change in spectral shape except change in intensity due to variation in Sm^{3+} ion concentration. This is because of lack of overlapping bands in the case of Sm^{3+} ions. Hence, no radiation trapping effects observed in these glasses due to the lack of overlapping bands. The observed results are purely due to the variation of Sm^{3+} concentrations but not on size and geometry of the Sm^{3+} glasses. The similar behaviour was observed in the case of lifetime of $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transitions at high concentration of Sm^{3+} ions. Moreover, multi-phonon relaxation, energy transfer and cross-relaxation channels were also stimulus the quenching of luminescence at high Sm^{3+} ion concentrations.

Partial energy level diagram of Sm^{3+} ions in GeBiNaBaGdSm glasses is presented in Fig. 4 with energy transfer (ET), multi-phonon relaxation (MPR), cross-relaxation (CR), non-radiation and radiation transitions. A blue excitation energy of 24691 cm^{-1} was used to pump the Sm^{3+} ions to the $^6\text{P}_{3/2}$ level. Because of the short lifetime of the $^6\text{P}_{3/2}$ level, Sm^{3+} ions rapidly de-excite to the level $^4\text{G}_{5/2}$ through non-radiative transitions. A large band gap with energy of 7210 cm^{-1} was perceived between the meta-stable state $^4\text{G}_{5/2}$ with its next lower

energy state $^6\text{F}_{11/2}$ which is caused to low non-radiative decay. Energy transfer due to cross-relaxation occurs between the energy levels from $^6\text{F}_J \rightarrow ^6\text{H}_J$ and $^6\text{H}_J \rightarrow ^6\text{F}_J$. Non-radiative emission takes place attributable to the Sm^{3+} ions relax from the $^6\text{P}_{3/2} \rightarrow ^4\text{G}_{5/2}$ transitions. Laser emission (radiative) can be obtained due to the de-excitation of Sm^{3+} ions from the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_J$ transitions. Because of closely space among many energy levels, multi-phonons can cause luminescence quenching at high Sm^{3+} concentrations. However, a large bandgap can reduce the multi-phonon relaxation in GeBiNaBaGdSm glasses. Due to the consideration of low phonon energy (880 cm^{-1}) of germanate glasses [39] nearly nine phonons are required to bridge the gap (7210 cm^{-1}) between the ground $^4\text{G}_{5/2}$ and excited $^6\text{F}_{11/2}$ states. As a consequence, multi-phonon relaxation can be ignored. Hence, the ET through CR channels is responsible for the quenching of luminescence.

The average integrated intensity ratio (red to orange, $\text{R/O} = I_{707}/I_{597}$) for GeBiNaBaGdSm glasses is found to be 0.13. The ratio of R/O intensities of $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ and $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{5/2}$ transitions is an assessment of the deviation from the centrosymmetric symmetries and is used to study the coordination environment around the Sm^{3+} ions. The intensity ratio mainly concentrates the bonding between ligand and Ln^{3+} ions. Smaller values of R/O indicate that a low deviation around the Sm^{3+} ion in the glass network. This disorder can be increased with the increase of Sm^{3+} ion concentration. Strong ionic bond nature of the Sm^{3+} ions with ligand is also responsible for obtaining low value of the R/O intensity ratio.

5.1. Radiative properties

Radiative properties that comprise of radiative transition probability A_R (s^{-1}), radiative lifetime τ_R (s) and branching ratio β_R of GeBiNaBaGdSm15 glass were estimated from the JO analysis. The magnitude of branching ratio can cause more lasing in the transition of emission. It is known that when the emission with branching ratio is more than 50% that could hold more potential for laser. The β_{exp} for $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition in GeBiNaBaGdSm15 glass is evaluated as 59%. The stimulated emission cross-section, $\sigma_{\text{SE}} \times 10^{-22} (\text{cm}^2)$ of GeBiNaBaGdSm15 glass is calculated by Eq. (3) and is compared with other reported Sm^{3+} glasses [26–36] in Table 4.

$$\sigma_{\text{SE}} = \frac{\lambda_p^4}{8\pi n^2 c} \times \frac{A_R}{\Delta\lambda_{\text{eff}}} \quad (3)$$

where λ_p is the peak wavelength, $\Delta\lambda_{\text{eff}}$ is the effective linewidth and n is the refractive index of the glass.

Comparison of radiative properties of GeBiNaBaGdSm15 glass with other reported glasses were presented in Table 5. The σ_{SE} for GeBiNaBaGdSm15 glass was found to be $15.59 \times 10^{-22} \text{ cm}^2$, which is greater compared with other reported glasses [26–36]. The $\sigma_{\text{SE}} \times \Delta\lambda_{\text{eff}}$ for GeBiNaBaGdSm15 glass is evaluated as $20.59 \times 10^{-28} \text{ cm}^3$ which is higher compared to $7.00 \times 10^{-28} \text{ cm}^3$ for BGGSm10 glass [27], $12.47 \times 10^{-28} \text{ cm}^3$ for ZABBSm03 glass [35], $14.11 \times 10^{-28} \text{ cm}^3$ for SNbKZFSm10 glass [31], $14.27 \times 10^{-28} \text{ cm}^3$ for NNS05 glass [33], $15.11 \times 10^{-28} \text{ cm}^3$ for ZPABSm10 glass [30] and $16.3 \times 10^{-28} \text{ cm}^3$ for ZnPbWTe10 glass [34] but less than $20.83 \times 10^{-28} \text{ cm}^3$ for NKBSm10 glass [29].

Table 4

Radiative properties: λ_p , $\Delta\lambda_{\text{eff}}$, A , τ_R , β and σ_{SE} of the $^4\text{G}_{5/2}$ excited level of Sm^{3+} ions in GeBiNaBaGdSm15 glass.

Transitions from $^4\text{G}_{5/2}$	λ_p	$\Delta\lambda_{\text{eff}}$ nm	A (s^{-1})	τ_R (ms)	σ_{SE} 10^{-22} cm^2	β		$\sigma_{\text{SE}} \times \Delta\lambda_{\text{eff}}$ 10^{-28} cm^3
						β_{exp}	β_{cal}	
$^6\text{H}_{5/2}$	561	9.37	22.78	43.89	1.27	0.05	0.07	1.19
$^6\text{H}_{7/2}$	597	13.21	307.32	3.25	15.59	0.59	0.41	20.59
$^6\text{H}_{9/2}$	644	11.31	121.95	8.20	9.78	0.23	0.41	11.06
$^6\text{H}_{11/2}$	707	18.45	72.83	13.7	5.21	0.14	0.10	9.61

Table 5Comparison of radiative properties of GeBiNaBaGdSm15 glass with other reported glasses for the $^4G_{5/2} \rightarrow ^6H_{7/2}$ transition.

Glass Matrix	Ref.	A (s ⁻¹)	β_{exp}	τ_R (ms)	τ_{exp} (ms)	σ_{SE} 10 ⁻²² cm ²	$\Delta\lambda_{\text{eff}}$ (nm)	$\sigma_{\text{SE}} \times \Delta\lambda_{\text{eff}}$ 10 ⁻²⁸ cm ³
GeBiNaBaGdSm15	[PM]	307	58	3.25	1.16 ± 0.01	15.59 ± 0.10	13.21 ± 0.01	20.59 ± 0.15
PKBFAS10	[26]	159	39	6.28	2.25	9.10	12.3	—
BGGSm10	[27]	—	36	2.91	0.99	4.84	14.46	7.00
PZSBSm15	[28]	453	35	2.20	—	9.13	18.3	—
NKBSm10	[29]	278	57	3.59	2.38	12.90	16.21	20.83
ZPABSm10	[30]	268	47	3.73	—	8.31	18.19	15.11
SNbKZFSm10	[31]	253	58	3.95	0.79	8.30	17	14.11
LGTSm10	[32]	296	46	3.37	0.79	9.20	—	—
NNS05	[33]	172	50	5.81	2.06	10.02	14.25	14.27
ZnPbWTe10	[34]	558	55	1.79	0.60	15.40	10.54	16.3
ZABBSm03	[35]	202	45	4.95	2.34	12.31	1.01	12.47
CASSm05	[36]	132	37	7.57	1.70	0.43	—	—

6. Luminescence decay profiles

6.1.

Photoluminescent decay curves of $^4G_{5/2}$ level of Sm^{3+} -doped GeBiNaBaGdSm glasses were recorded upon the excitation of 405 nm. The decay curve of GeBiNaBaGdSm005 glass reveal the exponential behaviour, as shown in Fig. 5 and is analysed by using a single exponential equation. The non-exponential behaviour was observed for the glasses from GeBiNaBaGdSm01 to GeBiNaBaGdSm25, which analysed by using a double exponential equation and Inokuti and Hirayama (I-H) model. The I-H model is used to fit the non-exponential behaviour of the glasses to understand the energy transfer process and multipolar interactions among Sm^{3+} ions. At high Sm^{3+} concentration, the decay profiles clearly demonstrate non-exponential behaviour which is owing to energy transfer among donors and acceptors of Sm^{3+} ions through non-radiative transitions. The ET is because of cross-relaxation directly depends on donor-acceptor separation (R_0) and donor-acceptor interaction parameter (C_{DA}) of Sm^{3+} ion concentration. The experimental lifetimes of the non-exponential decay curves were evaluated from equation (4)

$$\tau_{\text{exp}} = \frac{a_1 \tau_1^2 + a_2 \tau_2^2}{a_1 \tau_1 + a_2 \tau_2} \quad (4)$$

where τ_{exp} is the experimental lifetime, a_1 and a_2 are constants τ_1 and τ_2

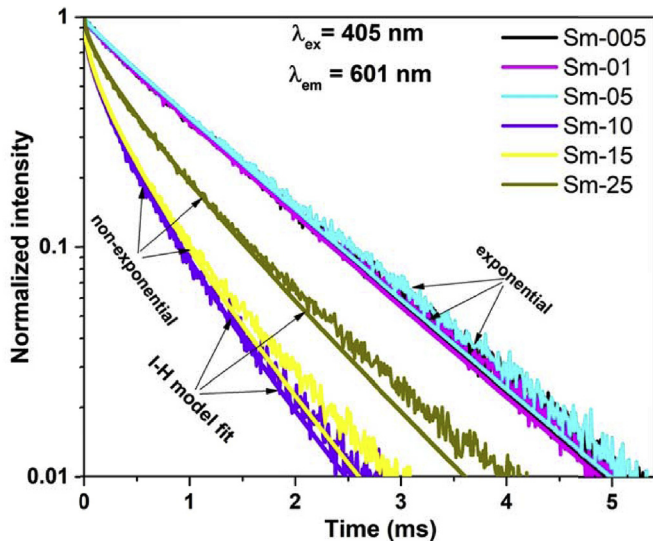


Fig. 5. Luminescence decay profiles of GeBiNaBaGdSm glasses with I-H model fit.

Table 6

Glass code, experimental lifetime and Q values of GeBiNaBaGdSm glasses.

Glass code	τ_{exp} (ms)	Q value
GeBiNaBaGdSm005	1.11	0.19
GeBiNaBaGdSm01	1.12	0.21
GeBiNaBaGdSm05	1.17	0.16
GeBiNaBaGdSm10	0.50	1.70
GeBiNaBaGdSm15	0.57	1.59
GeBiNaBaGdSm25	0.81	0.86

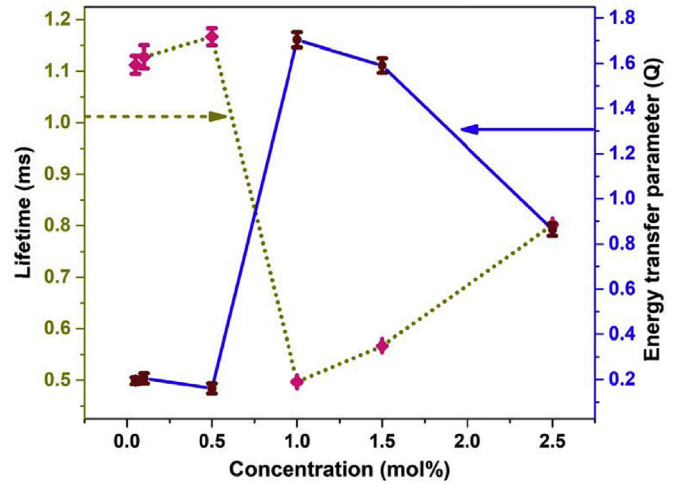


Fig. 6. The variation of lifetime and energy transfer parameter (Q) as function of Sm^{3+} concentration in of GeBiNaBaGdSm glasses.

are times of decay for exponential components.

The experimental lifetime (τ_{exp}) of the GeBiNaBaGdSm glasses is shown in Table 6. The variation of the lifetime and energy transfer parameter for different glasses is shown in Fig. 6. High lifetime value was reported to the GeBiNaBaGdSm05 glass, found to be 1.17 ms. At high Sm^{3+} ions concentration, the decrease in lifetime may be due to CR channels as shown in Fig. 4.

At low concentration of Ln^{3+} ions, the interactions among the ions are insignificant. As a result the decay curves can be fitted to single exponential function. On the other hand, at higher concentrations among the Ln^{3+} ions turn out to be so outstanding that the ET ensues from an excited Ln^{3+} ion (donor) to a ground state Ln^{3+} ion (acceptor), leading to a non-exponential decay curves. The donor to acceptor ET can be evaluated using Inokuti and Hirayama's (I-H) formula given in Eq. (6)

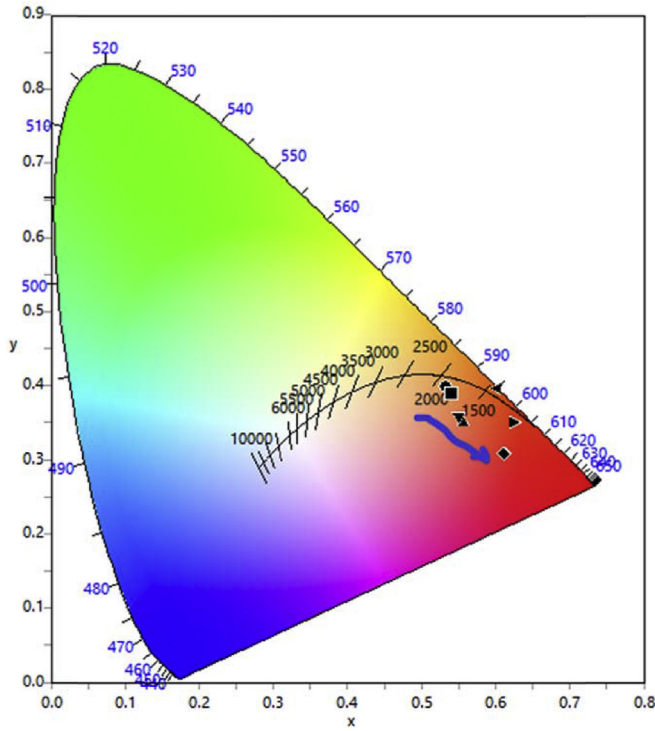


Fig. 7. The CIE chromaticity coordinates GeBiNaBaGdSm glasses.

$$I(t) = I_0 \exp \left[-\frac{t}{\tau_0} - Q \left(\frac{t}{\tau_0} \right)^{3/5} \right] \quad (6)$$

where t is the time after excitation, $\tau_0 = 1.11$ ms is the intrinsic decay time of donor in the absence of the acceptor. The interaction among ions whether dipole-dipole, dipole-quadrupole or quadrupole-quadrupole, respectively, can be governed by S ($= 6, 8, 10$). In Eq (6), Q is the ET parameter and is given by

$$Q = \frac{4\pi}{3} \Gamma \left(1 - \frac{3}{S} \right) N_0 R_0^3 \quad (7)$$

where Γ is the Euler function, N_0 is the concentration of acceptor ions and R_0 is the critical distance defined as donor-acceptor separation for which the rate of ET from donor to the acceptor is equal to the rate of intrinsic decay of the donor. The Γ is 1.77, 1.43 and 1.33 for the value of $S = 6, 8$ and 10 , respectively. Thus, the donor-acceptor interaction parameter C_{DA} is associated to the R_0 by Eq. (8).

$$C_{DA} = \frac{R_0^S}{\tau_0} \quad (8)$$

The luminescent decay profiles of GeBiNaBaGdSm glasses were successfully fitted by the I-H model for $S = 6$ and reveals that the interaction among Sm^{3+} ions are of dipole-dipole. The results suggest that the non-exponential behaviour of decay curves at higher concentrations of Sm^{3+} ions are owing to increase of energy transfer Q with increasing concentration of Sm^{3+} ions as shown in Fig. 7. The high ET was noticed for GeBiNaBaGdSm10 glass. The R_0 and C_{DA} of GeBiNaBaGdSm10 glass are found to be $10.28 \text{ \AA}$ and $9.73 \times 10^{-41} \text{ cm}^6/\text{s}$, respectively. The value of $C_{DA} (\times 10^{-41})$ of GeBiNaBaGdSm10 glass is comparable with 9.32 for PTBSm4 [40] and 9.43 for PKAZFSm30 [41], 7.30 for SNbKZFSm10 [31], 22.30 for BGGSm10 [27] and 1.16×10^{-42} for SNbKZFSm10 [29].

In Ln^{3+} ions doped glasses, τ_{exp} of the decay curves can be evaluated by the given formula:

$$\frac{1}{\tau_{\text{exp}}} = \frac{1}{\tau_{\text{cal}}} + W_{\text{MPR}} + W_{\text{ET}} \quad (9)$$

where W_{MPR} and W_{ET} are multi-phonon relaxation rate and energy transfer rate. The large energy gap of Sm^{3+} ions conquers the W_{MPR} . Quenching of lifetimes as a function of concentration is because of W_{ET} .

$$W_{\text{ET}} = W_{\text{CR}} = \frac{1}{\tau_{\text{exp}}} - \frac{1}{\tau_{\text{cal}}} \quad (10)$$

ET process is a significant factor in Sm^{3+} ions doped glasses that cause to increase in laser threshold by means of ion-ion interactions. The W_{ET} or W_{CR} for GeBiNaBaGdSm10 glass is found to be 554 s^{-1} compared to other reported glasses that includes 140 s^{-1} for NNS05 [33], 235 s^{-1} for CASSm05 [36], 396 s^{-1} for BGGSm10 [27], 436 s^{-1} for ZnPbWTe10 [34], 535 s^{-1} for NKBSm10 [29], 613 s^{-1} for LGTSm10 [32].

7. CIE chromaticity diagram

CIE chromaticity diagram was used for evaluating the color representation of the visible emission of Sm^{3+} -doped GeBiNaBaGdSm glasses. The chromaticity coordinates are found to be (0.53, 0.39), (0.54, 0.38), (0.56, 0.36), (0.56, 0.35), (0.55, 0.36), and (0.62, 0.35) for the Sm^{3+} concentration of 0.05, 0.1, 0.5, 1.0, 1.5 and 2.5 mol%, respectively, by following the procedure has been described elsewhere [42]. The locations of all the coordinates were shown in Fig. 7. The emission from the CIE diagram indicates that the high emission was observed at the orange region then shifting towards red as the Sm^{3+} ion concentration increases from 0.05 to 2.5 mol%.

8. Conclusion

Alkali bismuth germanate glasses doped with different Sm^{3+} ions concentration have been fabricated using melt quenching technique and studied their optical and spectroscopic properties. Nephelauxetic ratio and bonding parameters of the GeBiNaBaGdSm15 glass were determined and revealed the ionic bond nature of the Sm^{3+} ion-ligand. Judd-Ofelt (JO) parameters have been carried out to find the radiative properties for the excited states of Sm^{3+} ions in the GeBiNaBaGdSm15 glass. Upon 405 nm excitation intense orange-red emission was perceived due to the ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{7/2}$ transition. I-H model was employed to study non-exponential decay curves at higher concentrations of Sm^{3+} ions and revealed the existence of dipole-dipole interaction among Sm^{3+} ions. Energy transfer process was studied and C_{DA} and W_{CR} are determined which are of the order of $9.73 \times 10^{-41} \text{ cm}^6/\text{s}$ and 554 s^{-1} , respectively, for Sm^{3+} -doped GeBiNaBaGdSm10 glass. CIE diagram evidenced that the intense orange-red emission was obtained for GeBiNaBaGdSm glasses that could be useful for visible display device applications.

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